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A hybrid learning approach for MRI-based detection of alzheimer's disease stages using dual CNNs and ensemble classifier

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Alzheimer's Disease (AD) and related dementias are significant global health issues characterized by progressive cognitive decline and memory loss. Computer-aided systems can help physicians in the early and accurate detection of AD, enabling timely intervention and effective management. This study presents a combination of two parallel Convolutional Neural Networks (CNNs) and an ensemble learning method for classifying AD stages using Magnetic Resonance Imaging (MRI) data. Initially, these images were resized and augmented before being input into Network 1 and Network 2, which have different structures and layers to extract important features. These features were then fused and fed into an ensemble learning classifier containing Support Vector Machine, Random Forest, and K-Nearest Neighbors, with hyperparameters optimized by the Grid Search Cross-Validation technique. Considering distinct Network 1 and Network 2 along with ensemble learning, four classes were identified with accuracies of 95.16% and 97.97%, respectively. However, using the derived features from both networks resulted in an acceptable classification accuracy of 99.06%. These findings imply the potential of the proposed hybrid approach in the classification of AD stages. As the evaluation was conducted at the slice-level using a Kaggle dataset, additional subject-level validation and clinical testing are required to determine its real-world applicability.

Keywords Alzheimer's disease staging, Magnetic resonance imaging, Decision support system, Deep learning, Ensemble learning

Alzheimer's Disease (AD) is a neurodegenerative disorder that predominantly affects the elderly population. It is characterized by the deterioration of cognitive and functional abilities, which eventually leads to death[1]. First documented by Alois Alzheimer in 1906, AD was initially identified in a 51-year-old woman who exhibited symptoms of cognitive disturbances, disorientation, delusions, and other behavioral changes[2]. Pathologically, the accumulation of beta-amyloid plaques and tau tangles in the brain destroys neurons and causes an irreversible loss of brain cells[3,4]. As the disease progresses, multiple brain regions and networks that are fundamental for thinking, memory, planning, and decision-making undergo atrophy[5].

In recent years, AD has become a significant public health concern. It is estimated that by the end of 2025, one in every 85 individuals may suffer from this illness[6,7]. Because AD has a progressive nature, early diagnosis is critical for implementing effective management strategies. However, diagnosing AD in its early stages is difficult due to the subtlety of initial symptoms and the complexity of identifying definitive biomarkers. When it comes to diagnosing AD, psychological tests and medical imaging techniques, such as Magnetic Resonance Imaging (MRI) and Functional Magnetic Resonance Imaging (fMRI), provide valuable insights[8,9]. Neuroimaging methods not only assist in diagnosing the most common neurodegenerative conditions after the onset of symptoms but also hold diagnostic potential in the very early or presymptomatic phases of the disease[10]. While these techniques have improved diagnostic precision, their application to large populations of AD patients requires substantial financial and time investments, posing challenges for widespread implementation.

Regarding the intricacy of information, medical professionals are required to carefully analyze patients' data. Therefore, developing an automatic detection and classification method is necessary to streamline the identification and categorization of complex medical data[11]. Utilizing machine learning and deep learning algorithms enables the extraction of high-level features, allowing for rapid and cost-effective recognition of diseases and their progressive stages through classification frameworks[12,13]. Numerous studies have focused

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on the automatic detection of AD. For instance, Arafa et al.[14]. developed a framework using neural networks and transfer learning for the diagnosis and classification of AD. Their methodology involved pre-processing, data augmentation, feature extraction, and classification stages. The proposed Convolutional Neural Network (CNN) model demonstrated high diagnostic precision with an accuracy of 99.95%, while the VGG network, a type of very deep CNN, attained an accuracy of 97.44%. Savaş et al.[15]. employed diverse CNN models to classify 2182 image objects sourced from the ADNI database. Among the pre-trained models evaluated, the EfficientNet achieved the highest classification accuracy, demonstrating its effectiveness in distinguishing AD-related images. Antony et al.[16]. proposed two models, VGG16 and VGG19, for diagnosing AD. These models were trained on the 780-image ADNI dataset. During the pre-processing stage, the skull was removed, and data augmentation techniques were applied. Both models had adequate accuracy, with VGG-16 at 81% and VGG-19 at 84%. Hosseini-Asl et al.[17]. introduced a deep 3D-CNN design to learn universal and transferable features across various domains. This model was capable of identifying the characteristic AD biomarkers in one (source) domain and carrying out task-specific classification in another (target) domain. Farooq et al.[18]. introduced a CNN-based pipeline for diagnosing AD and its stages using MRI images. Liu et al.[19]. developed a cascaded CNN structure to learn the multi-level and multimodal features of Positron Emission Tomography (PET) and MRI images for AD diagnosis. In another study, Yildirim et al.[20]. analyzed AD classification using a four-class dataset. They used CNN architectures and compared their results with a proposed hybrid model based on ResNet50. The hybrid model achieved an accuracy rate of 90%, outperforming pre-trained CNN models. The results proved the robustness of hybrid models in order to overcome the challenges of AD detection.

This study introduces a novel dual-stream CNN design for feature extraction from MRI images across various stages of AD, which differs from conventional deep networks. Rather than stacking many layers within a single pathway, two parallel, low-complexity CNNs with distinct architectural traits were employed, allowing each stream to specialize in extracting different types of features. In both networks, an attention layer was used to enhance feature relevance. Features extracted from both CNN branches were then fused and fed into a soft-voting ensemble of optimized classifiers to identify different stages of AD.

Materials and methods

The overall steps of the proposed method, including pre-processing, feature extraction, and classification, are illustrated in Fig. 1. According to the figure, MRI images were resized after normalization, split into training and test sets, and the Synthetic Minority Over-sampling Technique (SMOTE) was applied for data augmentation and balancing class distributions in the training data. Complex features were then extracted through parallel CNNs (Network 1 and Network 2) and subsequently fused. The selected features were fed into an ensemble learning with soft voting, combining Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN) classifiers to achieve final classification accuracy across Alzheimer's stages, with hyperparameters optimized via the Grid Search Cross-Validation (GridSearchCV) method.

Data description and pre-processing

The dataset of this study is a publicly accessible Alzheimer's disease dataset sourced from Kaggle[21]. It contains 6400 T1-weighted MRI images and the corresponding class label information of 200 anonymous patients, in which each subject has 32 horizontal brain slices. These images were categorized into four states: non-demented (ND), very mild demented (VMD), mild demented (MID), and moderate demented (MOD). All images were resized from their original dimensions of 208×176 to 176×176 prior to normalization. Then, the dataset was divided into a training set (80%) and a test set (20%). Since there were unbalanced images corresponding to disease progression, the SMOTE method was applied exclusively to the training sets to produce synthetic data for each imbalanced category. Finally, the dimensions of the images were adjusted to $176 \times 176 \times 1$ to meet the input requirements of the CNN architecture. Table 1 details the number of total images, as well as the distribution between training and test sets for each class. It also includes the number of training images after applying the SMOTE technique.

Feature extraction

In the realm of neural networks, CNNs play a significant role in the extraction of unique features, especially in the domain of deep learning[22]. The input layer of this network is typically an image of arbitrary dimensions, and its output is a high-resolution vector representing different classes. The hidden layer consists of the convolutional layer, the pooling layer, and the fully connected layer[23]. The convolutional layers are responsible for filtering the input data through various kernels, allowing the network to capture important patterns and features[24]. Meanwhile, the pooling layers help reduce the dimensionality of the data, and the fully connected layer incorporates high-level features.

In this manuscript, to effectively extract the range of shallow and deep features for classifying AD progression, the dual-CNNs were designed. Tables 2 and 3 summarize the architecture of each CNN, referred to as Network 1 and Network 2, respectively. Network 1 contains three Conv2D layers with ReLU activation functions. In the structure of this network, max-pooling layers for down-sampling, and an attention layer to enhance the focus on critical features were used. Furthermore, batch normalization and dropout techniques were employed between layers to regularize and enhance training effectiveness. Network 2 also followed a similar structure with minor differences in the sequence of layers and the pooling method. It incorporated average-pooling in the earlier layers, and an attention layer at deeper stages. In the fusion network, the flattened features from Network 1 (3,872 features) and Network 2 (3,387 features) were concatenated to form a combined feature vector. This fused vector was then passed through consecutive dense layers to reduce its dimensionality: first to 512 features, and then to 128 features. It is important to note that a feedforward approach was utilized to ensure the sequential flow of data from the input layer to the dense layers. To optimize the performance of networks,

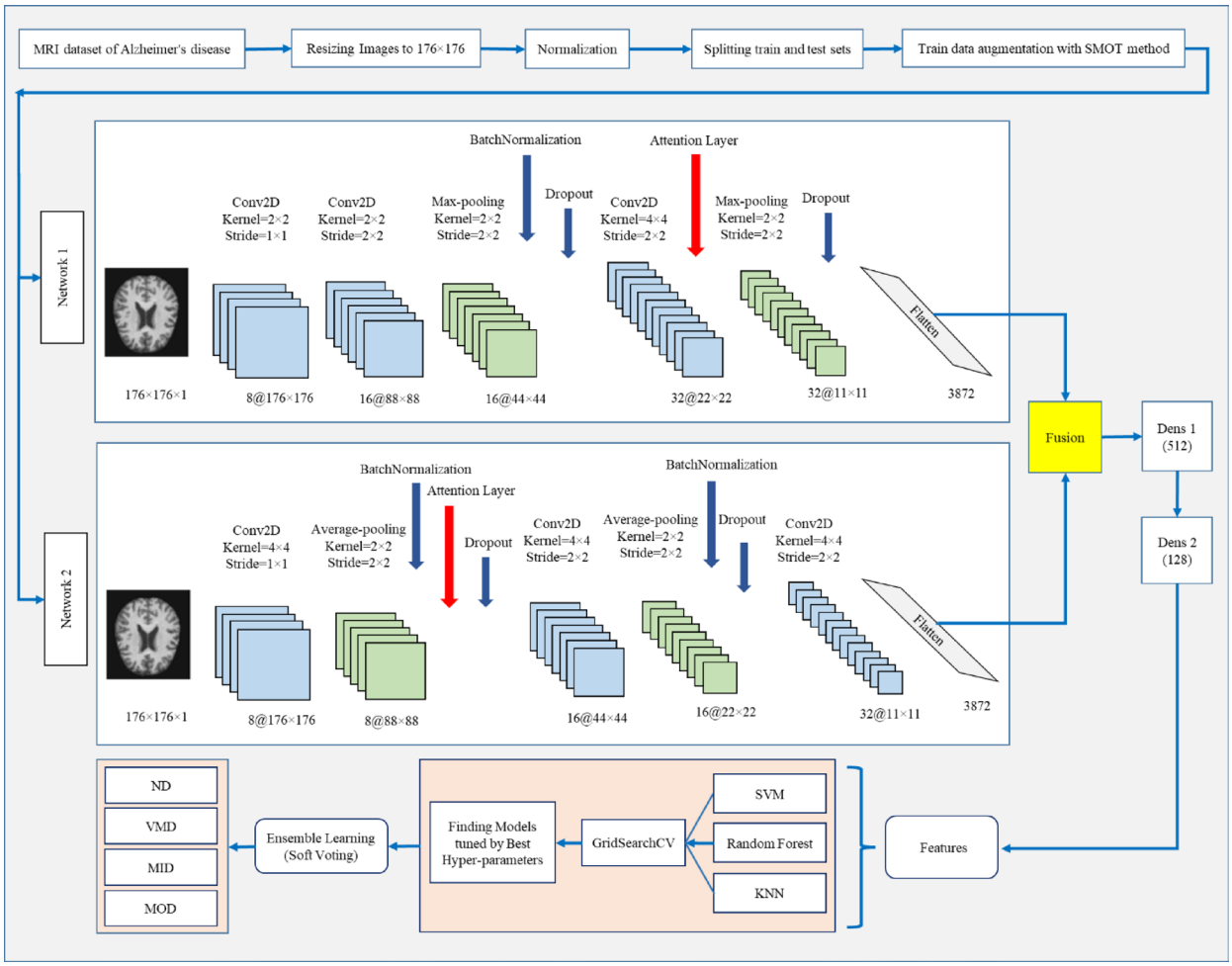


Fig. 1. The framework of the proposed method, including pre-processing, feature extraction via a parallel CNN, and classification of AD stages using an ensemble learning method consisting of SVM, RF, and KNN classifiers optimized by GridSearchCV.

No.	Class		Test images	Train images	Train images after augmentation
1	non-demented (ND)	3200	640	2560	2560
2	very mild demented (VMD)	2240	448	1792	2560
3	mild demented (MID)	896	180	716	2560
4	moderate demented (MOD)	64	13	51	2560

Table 1. Number of MRI images sourced from the kaggle dataset, including totals, training and test sets, and training set after data augmentation.

backpropagation was employed as the learning algorithm. It involves the computation of gradients of the loss function with respect to each weight by applying the chain rule, allowing for efficient updates during training. The training process of networks was further refined using the Adam optimizer with a learning rate of 0.001, a batch size of 100, and 38 epochs.

Classification

For classification, the ensemble method containing multiple machine learning models (SVM, KNN, and RF) was employed to improve prediction accuracy, as shown in Fig. 2. SVM is a supervised learning method that identifies the optimal hyperplane, maximizing the margin between classes[25]. This method can effectively classify both binary and multi-class data in a high-dimensional space with the help of a kernel function[26]. KNN, as a non-parametric algorithm, determines the class of the test samples based on the majority vote of its k-nearest neighbors. In contrast, the RF technique constructs several decision trees during training and combines their outputs through voting for prediction improvement as well as for overfitting reduction.

Layer (Type)	Units/Kernel size	Stride	Activation function	Padding	Output shape	Parameters
Input	-	-	-	-	176 × 176 × 1	0
Conv2D	8 filters, (2 × 2)	1 × 1	ReLU	Same	176 × 176 × 8	40
Conv2D	16 filters, (2 × 2)	2 × 2	ReLU	Same	88 × 88 × 16	528
Max-Pooling	2 × 2	2 × 2	-	-	44 × 44 × 16	0
Batch-normalization	-	-	-	-	44 × 44 × 16	64
Dropout (0.4)	-	-	-	-	44 × 44 × 16	0
Conv2D	32 filters, (4 × 4)	2 × 2	ReLU	Same	22 × 22 × 32	8224
Attention Layer	1 filter, (1 × 1)	1 × 1	Sigmoid	valid	22 × 22 × 32	32
Max-Pooling	2 × 2	2 × 2	-	-	11 × 11 × 32	0
Dropout (0.4)	-	-	-	-	11 × 11 × 32	0
Flatten	-	-	-	-	3872	0
Dense 1	-	-	-	-	512	1,984,896
Dense 2	-	-	-	-	128	65,664

Table 2. The detailed architecture of network 1 employed in the proposed method.

Layer (Type)	Units/Kernel size	Stride		Padding	Output shape	Parameters
Input	-	-	-	-	176 × 176 × 1	0
Conv2D	8 filters, (4 × 4)	1 × 1	ReLU	Same	176 × 176 × 8	136
Avg-Pooling	2 × 2	2 × 2	-	-	88 × 88 × 8	0
Batch-normalization	-	-	-	-	88 × 88 × 8	32
Attention Layer	1 filter, (1 × 1)	1 × 1	Sigmoid	valid	88 × 88 × 8	9
Dropout (0.4)	-	-	-	-	88 × 88 × 8	0
Conv2D	16 filters, (4 × 4)	2 × 2	ReLU	Same	44 × 44 × 16	2064
Avg-Pooling	2 × 2	2 × 2	-	-	22 × 22 × 16	0
Batch-normalization	-	-	-	-	22 × 22 × 16	64
Dropout (0.4)	-	-	-	-	22 × 22 × 16	0
Conv2D	32 filters, (4 × 4)	2 × 2	ReLU	Same	11 × 11 × 32	8224
Flatten	-	-	-	-	3872	0
Dense 1	-	-	-	-	512	1,984,896
Dense 2	-	-	-	-	128	65,664

Table 3. The detailed architecture of network 2 employed in the proposed method.

In the classification stage, the same images selected for the training set (80%) and test set (20%) during the pre-processing step were used to avoid any contribution of augmented data in the test set. Hyperparameter tuning was performed using 10-fold cross-validation with grid search on the training set, optimizing the regularization parameter for SVM (C = 1, 3, 5), the number of neighbors for KNN (K = 1, 3, 5), and the number of estimators for RF (N = 30, 40). Since the best-performing models consistently had optimal values clustered around specific hyperparameters during an initial broad grid search, narrower hyperparameter ranges were selected to reduce computational costs. All other hyperparameters were maintained at their default values, as specified in the scikit-learn library in Python. Based on the cross-validation performance, the best-performing model for each algorithm was obtained to predict the labels of the test set[27]. Eventually, the soft voting ensemble integrated the models' strengths by averaging the predicted probabilities from each model to create a consensus-driven final prediction[28].

Results and discussion

Performance setup

The proposed method was simulated in the Spyder environment (Python 3.8). The performance of this method for classifying AD progression is illustrated in Fig. 3, which provides a clear comparison between the effectiveness of networks in extracting features and classifiers in distinguishing four states: ND, VMD, MID, and MOD. The results indicate that utilizing features extracted from Network 1 for classification led to moderate accuracies compared to Network 2, with a slight improvement of nearly 3% across all classifiers. Nevertheless, the fused strategy that combines features from two networks improved results, with accuracies of 96.64% with CNN, 98.36% with RF, 98.83% with SVM, 99.45% with KNN, and 99.06% using ensemble learning with a soft-voting technique. It is clear that the KNN classifier and the ensemble learning method achieved the highest classification accuracies in comparison with RF and SVM. Among these classifiers, using ensemble learning seems to be more reliable and robust for classification tasks. To examine the contribution of attention layers in the suggested dual-

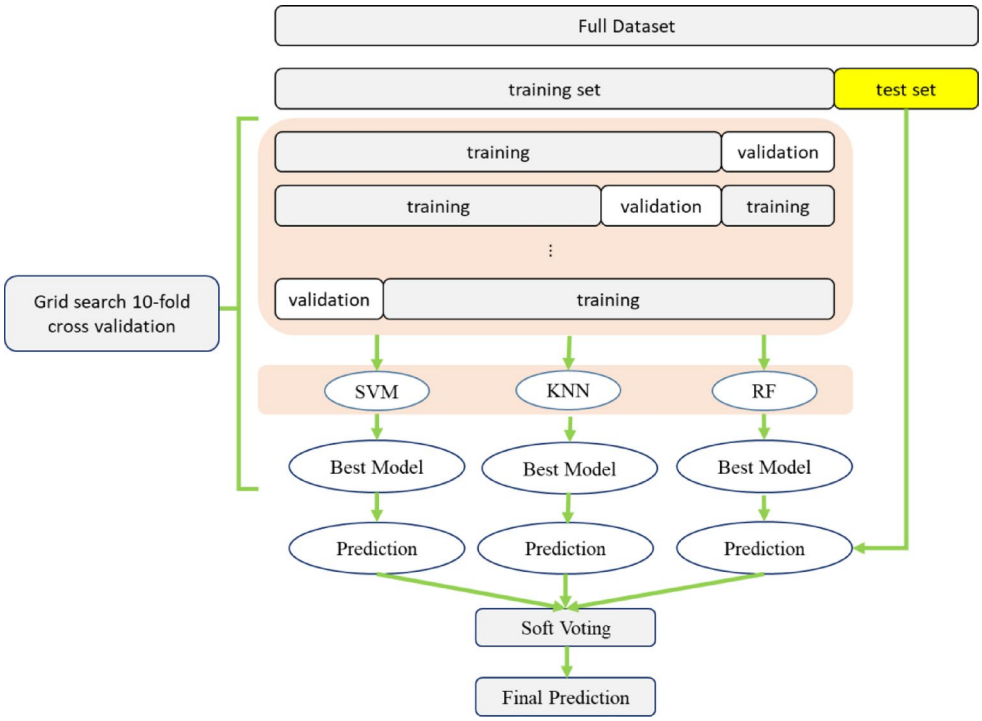
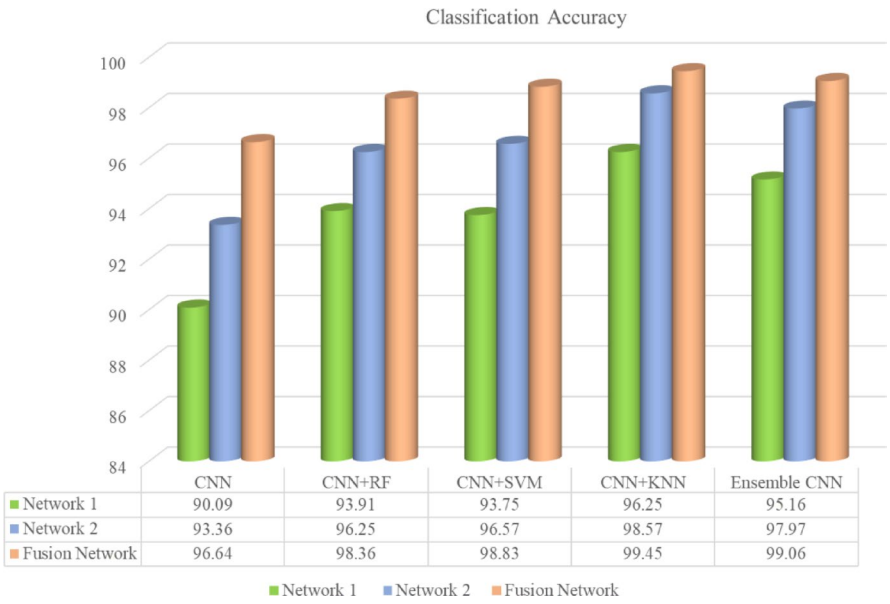


Fig. 2. Grid search-based ensemble learning architecture with cross-validation for AD classification.



	CNN	CNN+RF	CNN+SVM	CNN+KNN	Ensemble CNN
Network 1	90.09	93.91	93.75	96.25	95.16
Network 2	93.36	96.25	96.57	98.57	97.97
Fusion Network	96.64	98.36	98.83	99.45	99.06

Fig. 3. The comparison of classification accuracy utilizing features extracted by various network models.

network model, these layers were removed, and the model's performance was re-evaluated. Without attention layers, the classification accuracy fell to 96.57% (RF), 97.27% (SVM), 97.97% (KNN), and 97.74% (ensemble learning). Decreasing the results by approximately 2% demonstrates that considering attention layers in the CNN structure can have a positive impact on the feature extraction process.

Additionally, the training process, including CNN training, feature extraction from the training set, and training the ensemble classifier, lasted for 1,703, 1,734, and 2,662 s with Network 1, Network 2, and the fusion network, respectively. In contrast, feature extraction from the test set and label prediction were completed in less than 2 s. The rapid computation time during the inference phase makes this approach suitable for clinical application use.

	Classes	precision	recall	F1-score	Accuracy
Network 1	ND	95.65	96.25	95.95	95.94
	VMD	93.73	93.52	93.63	95.55
	MID	96.61	95.00	95.79	98.82
	MOD	100.0	100.0	100.0	100.0
Network 2	ND	98.28	98.75	98.51	98.51
	VMD	97.32	97.54	97.43	98.20
	MID	98.29	96.11	97.19	99.21
	MOD	100.0	100.0	100.0	100.0
Fusion Network	ND	98.91	99.38	99.14	99.14
	VMD	99.10	98.44	98.76	99.14
	MID	99.44	99.44	99.44	99.84
	MOD	100.0	100.0	100.0	100.0

Table 4. Performance indices of classes according to feature extraction methods (Network 1, network 2, and fusion network) and ensemble learning classifier.

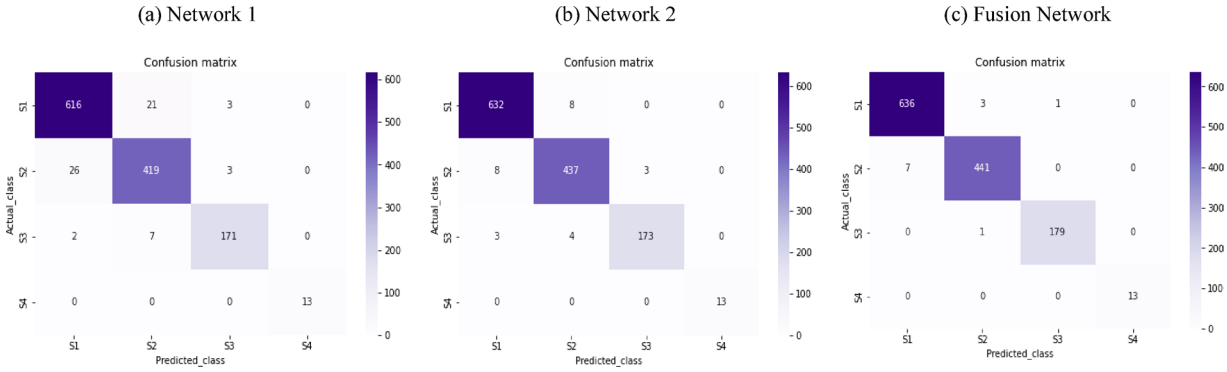


Fig. 4. Confusion matrices for the ensemble learning method considering features extracted from (**a**) Network1, (**b**) Network2, and (**c**) the fusion network. Rows indicate predicted classes and columns represent ground truth with ND, VMD, MID and MOD classes defined as S1, S2, S3, and S4, respectively.

For further analysis, the precision, recall, F1-score, and accuracy for each class were calculated and reported in Table 4, based on the derived features from each network and feeding them into an ensemble learning approach. It can be seen that acceptable results were obtained for both Network 1 and Network 2, particularly in identifying the MOD class; however, the fusion network outperformed them. Moreover, the combination of CNNs and the ensemble learning method could successfully handle the challenge of classifying moderate classes, such as VMD and MID, that share similarities. According to Network 1, the F1-score for the ND, VMD, MID, and MOD classes was accounted for 95.95%, 93.63%, 95.79%, and 100%, respectively. These scores notably increased to 98.51%, 97.43%, 97.19%, and 100% for Network 2. Although both networks identified the MOD class without error, some misclassifications occurred in the VMD and MID states and caused relatively low results. The use of a fusion network reduced classification errors in a way that F1-scores for these states became 98.76% and 99.44%, in the same order.

The confusion matrices in Fig. 4 represent the performance of the ensemble learning model in classifying four states, which uses features extracted and selected by various networks. While the MOD class was predicted without error by considering three networks, the fusion network demonstrated fewer misclassifications and provided more accurate predictions across all classes.

Moreover, Fig. 5 illustrates the Receiver Operating Characteristic (ROC) curves with the corresponding Area Under the Curve (AUC) values for multiclass classification. Panels (a), (b), and (c) related to features extracted by Network 1, Network 2, and Fusion Network, respectively, with classification performed using ensemble learning. Since the Fusion network achieved superior AUC values in all classes, the learning process of the fusion network through its loss and accuracy curves was visualized in Fig. 6. The loss curve exhibits a steady decrease over epochs, indicating error minimization during pattern learning. Conversely, a gradual increase can be seen in the accuracy curve over epochs, reflecting improvement in correct pattern identification. The negligible fluctuations during training suggest good generalization, stable convergence, and the effectiveness of the fusion network in extracting discriminative features.

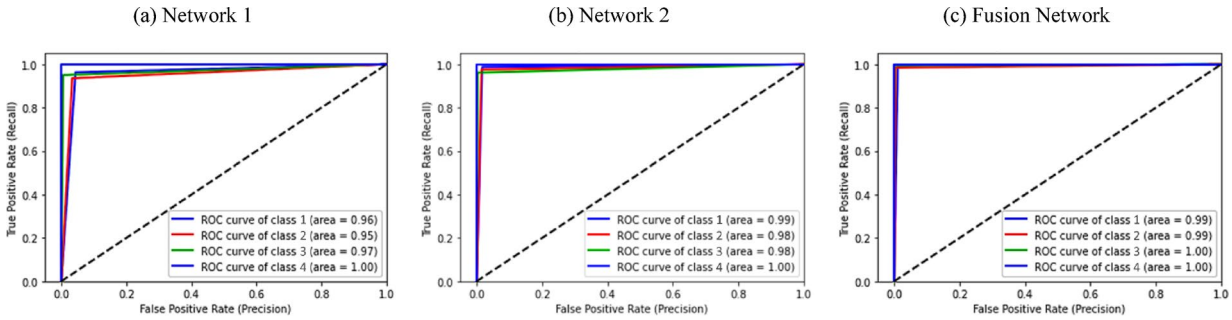


Fig. 5. ROC curves and corresponding AUC values for multiclass classification using ensemble learning, based on features extracted by (**a**) Network 1, (**b**) Network 2, (**c**) Fusion Network. The ND, VMD, MID, and MOD classes are defined as class1, class2, class3, and class4, respectively.

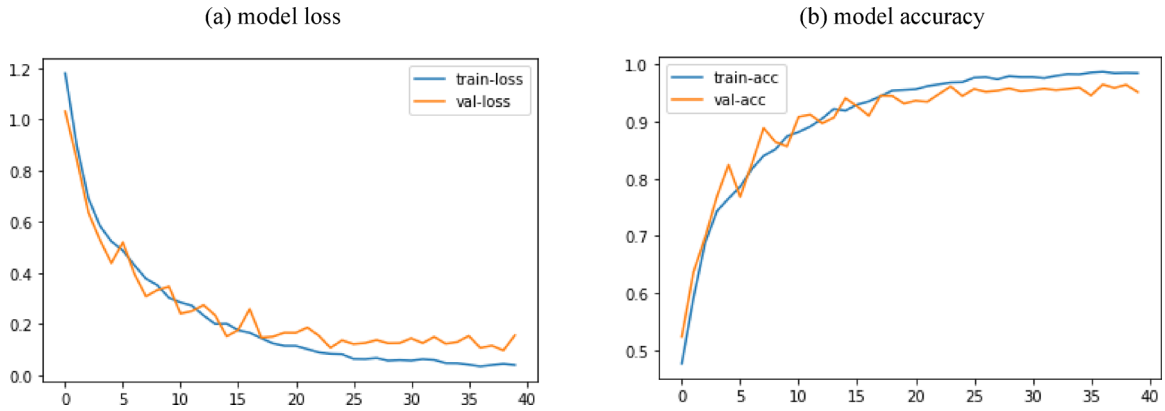


Fig. 6. Learning process of fusion network (**a**) model loss and (**b**) model accuracy.

Author(s)	Dataset	Classes	Methodology	Attention	F1-score (%)	Accuracy (%)
Al-Adhaileh[29].	Kaggle	4	AlexNet	No	94.12	94.53
Assmi et al.[30]	Kaggle	4	VGG-19	No	78.50	92.86
Murugan et al.[31]	Kaggle	4	SMOTE + DEMNET	No	95.50	95.23
Sahid et al.[32]	Kaggle	4	Federated Learning + CNN	No	84.00	84.75
El-Latif et al.[33]	Kaggle	4	Lightweight CNN	No	95.90	95.93
Al Rahbani et al.[34]	Kaggle	4	ResNet50 + EfficientNet	No	99.49	99.41
Balasundaram et al.[35]	Kaggle	4	Deep CNN + Ensemble learning	No	95.50	94.10
Chen et al.[13]	ADNI	2	3D Residual RepVGG Attention network	Yes	-	89.2
Islam et al.[36]	OASIS	4	Ensemble Deep CNN	No	92.00	93.18
Xin et al.[37]	OASIS	2	Efficient Conv-Swin Net + MLP	Yes	-	93.9
Hussain et al.[38]	OASIS	4	Transfer learning (AlexNet, GoogleNet, MobileNetV2)	No	-	98.2
The proposed method	Kaggle	4	Dual CNNs + attention layer + Ensemble classifier	Yes	99.33	99.06

Table 5. Performance analysis of the proposed method with other works.

Comparison with other models

We compared the proposed dual-CNN with an ensemble learning strategy to the most current state-of-the-art models for classifying AD stages using MRI data to put its performance and innovation in context. These comparisons are based on publicly accessible datasets like ADNI, OASIS, and Kaggle. The focus is on models created in recent years that utilize CNNs, attention mechanisms, and fusion methods. The methodological choices and performance metrics (accuracy and F1-score) of recent studies were summarized in Table 5.

Based on the proposed method, an acceptable accuracy of 99.06% was obtained on the Kaggle dataset for the classification of AD stages. This result surpasses other CNN-based and ensemble learning methods, including those without attention mechanisms. Notably, many of the baseline models use complex architectures, such as VGG-19, AlexNet, and ResNet, while higher accuracy was obtained by a lighter-weight CNN model. In this

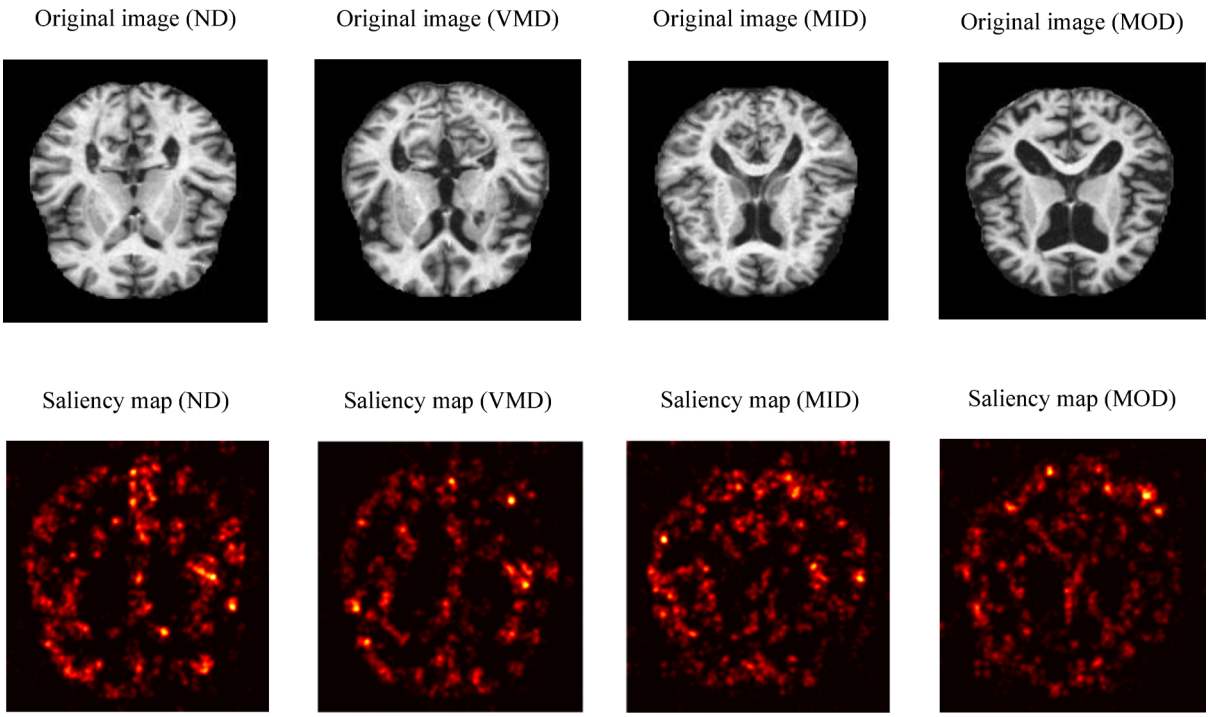


Fig. 7. Original MRI images of ND, VMD, MID, and MOD, along with their saliency maps obtained from the combination of features extracted and fused by the two branches of the CNN.

method, several steps were taken to enhance classification accuracy and solve some limitations of previous works. First, to handle class imbalance and avoid bias in prediction, the SMOTE method was applied to the training set. Second, the fusion of features derived from two lightweight CNN networks with different structures captured a richer and more diverse set of features suitable for distinguishing AD stages. In these networks, the inclusion of attention layers enhanced the focus on critical features that had been overlooked by some prior methods. Third, using ensemble learning with multiple classifiers (SVM, KNN, and RF) yielded robust results and performed better than methods that relied on a single classifier. Fourth, instead of using fixed or suboptimal hyperparameters, the GridSearchCV approach was employed to tune hyperparameters, ensuring that the final model was well-tuned. In summary, the integration of dual lightweight CNNs, attention, SMOTE, and ensemble classification with tuning distinguishes this method from prior works that typically adopt only one or two of these strategies.

Saliency maps and physiological discussion

As illustrated in Fig. 7, the first row displays the original MRI images of different states (ND, VMD, MID, and MOD). In the next row of this figure, the corresponding saliency maps highlight the most influential regions, indicated in red. Regarding the original images, healthy subjects have normal gray matter density without noticeable atrophy. However, signs of atrophy can be seen in the hippocampus, temporal regions, and entorhinal cortex of patients with VMD and MID. These changes can be associated with cognitive and memory impairments. In MOD images, the atrophy is conspicuous in various parts of the brain.

To evaluate the relevance of brain regions in the prediction process, saliency maps were created by calculating the gradient of the model's output with respect to the input images[39]. These maps provide a visual representation of the brain regions that the dual-CNN model primarily focuses on when distinguishing AD stages. As the disease progresses, saliency maps show a growing concentration and widespread range of red points. It is important to note that these maps were included to interpret the CNN's decision-making process, not for clinical use. They highlight regions the model finds important for classifying Alzheimer's disease stages, but these do not necessarily correspond to anatomically or physiologically meaningful areas.

Clinical relevance and applicability

The proposed dual-CNN and ensemble learning approach can serve as a clinical decision support system (DSS) to help radiologists and neurologists in classifying AD stages. Although this method demonstrated high slice-level classification accuracy on MRI data, clinical viability depends on further validation. In real-world settings, several factors such as variations in MRI acquisition protocols, scanner hardware, and patient demographics can impact performance. Additionally, deployment would require subject-level predictions based on 3D MRI volumes and patient metadata, not just 2D slices. In order to overcome these challenges and achieve more generalizable results, conducting further multicenter studies, developing domain adaptation techniques,

and integrating the model into clinical workflows is necessary. The ultimate goal is to improve DSS, enhance efficiency, and reduce the level of uncertainty in early-stage AD assessment.

Conclusion

In this work, a dual-CNN structure combining the outputs of Network 1 and Network 2 was utilized to extract salient features from MRI images, along with an ensemble learning method to perform AD classification. The evaluation of this model on the test data achieved an overall accuracy of 99.06%. In comparison, using Network 1 and Network 2 solely for feature extraction resulted in accuracies of 97.97% and 95.16%, respectively. Additionally, choosing ensemble learning as a classifier delivered better performance than the SVM, RF, and Softmax classifiers.

There are several limitations that need to be acknowledged. First, the architecture of dual-CNNs was constructed and evaluated based on a relatively small dataset, with particularly few samples in the MOD class. Although the SMOTE method was applied to solve the class imbalance issue, synthetic oversampling may increase the risk of overfitting. Second, this article focused solely on MRI images; it would be beneficial to investigate other imaging modalities in addition to this one. Third, the evaluation was carried out at the slice level due to the structure of the Kaggle dataset, which organizes 2D MRI slices into four folders corresponding to the AD stage. As a result, during train-test splitting, slices from the same subject may appear in both sets, which may cause data leakage. To address this limitation, future research will focus on datasets with subject-level annotations and entire 3D volumes to ensure more robust and clinically relevant validation. Moreover, further comparison will be made with other state-of-the-art models that integrate different deep learning and machine learning methods. Apart from MRI images, multimodal datasets comprising PET scans, EEG signals, and genetic biomarkers will be processed to enhance the reliability of AD classification.

Data availability

https://www.kaggle.com/datasets/marcopinamonti/alzheimer-mri-4-classes-dataset.

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Author contributions

SZ designed the methodology, conducted the simulation, and wrote the manuscript. AJ contributed to writing the manuscript and gathering information. YS also supervised the overall work. The authors reviewed and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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